



When a fast aircraft travels, it generates a series of pressure waves in the air. These waves travel at the speed of sound, approximately 761 mph (1,225 km/h) at sea level, and a little lower at higher altitudes. As the aircraft's speed increases, the waves are forced together to form a single shock wave, which makes a thunder-like boom when released. Most sonic booms

last between 0.1 and 0.5 seconds. The first supersonic flight was in 1947. Today, many military jet aircraft regularly travel at supersonic speeds. The F-14 has a top speed of more than 1,500 mph (2,400 km/h) at high altitude. Only two passenger airliners have ever operated at supersonic speeds: the Russian Tupolev Tu-144 and the British/French Concorde.